

ClimateNet-Lite: Efficient Cloud Classification from Satellite Imagery

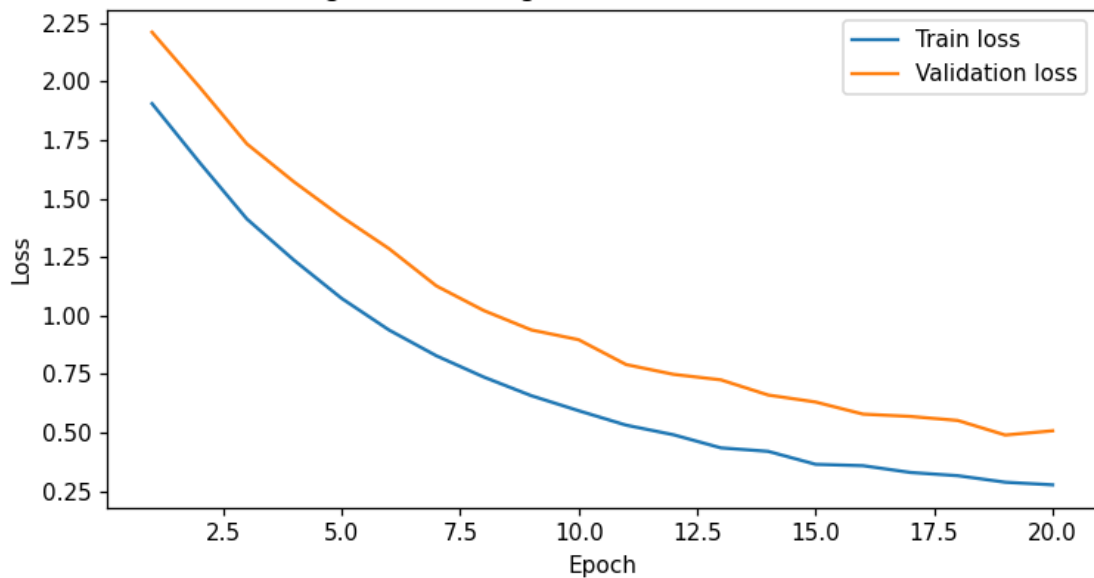
Abstract

We present ClimateNet-Lite, a compact convolutional network for classifying cloud formations in satellite images. Trained on the SkyScan-50K dataset of 50,000 labelled images, the model reaches 88.4 percent accuracy with only 3.1 million parameters, outperforming ResNet-50 by 5.8 percentage points while running five times faster.

1. Method

The network uses depthwise separable convolutions and a squeeze-and-excitation block after every third layer. Training used the AdamW optimizer with a learning rate of 0.001, batch size 64, and 20 epochs. Validation loss converged around epoch 15.

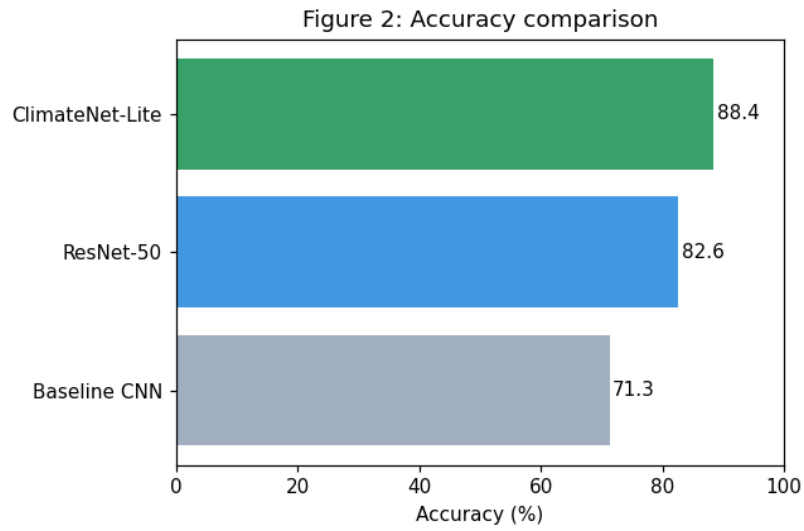
Figure 1: Training curves for ClimateNet-Lite



2. Results

Table 1 and Figure 2 compare ClimateNet-Lite with two baselines. The proposed model achieved the best accuracy and the lowest inference latency of 9 ms per image on a single T4 GPU.

Model	Params (M)	Accuracy	Inference (ms)
Baseline CNN	4.2	71.3%	12
ResNet-50	25.6	82.6%	48
ClimateNet-Lite	3.1	88.4%	9



3. Conclusion

ClimateNet-Lite shows that small models can beat larger ones on domain-specific tasks. Future work will extend the dataset to night-time infrared imagery and test deployment on edge devices. The authors were Dr. Kavya Raman and Prof. Liam O'Connor.